FINAL PROJECT REPORT

Noise Removal in Medical Images

Remove Noise While Preserving Edges in MRI or CT Images

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1. Abstract

This report presents a comprehensive evaluation of five image denoising methods applied to brain MRI images. The study addresses the critical problem of noise corruption in medical images, which can obscure anatomical features and hinder accurate clinical diagnosis. Two noise types were investigated: Gaussian noise (simulating MRI acquisition noise) at three severity levels ($\sigma = 10, 20, 30$), and salt-and-pepper noise (simulating CT detector artifacts) at two densities ($d = 2\%, 5\%$).

The five methods evaluated were: Median Filter, Gaussian Filter, Bilateral Filter, Non-Local Means (NLM), and a deep learning approach using DnCNN (Deep CNN for denoising). Performance was measured using Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Canny edge preservation analysis. Results demonstrate that no single method dominates across all noise types: Median Filter achieved the best performance against salt-and-pepper noise (PSNR = 35.03 dB, SSIM = 0.9739), while DnCNN outperformed all methods for heavy Gaussian noise (PSNR = 28.47 dB, SSIM = 0.7373 at $\sigma = 30$).

2. Introduction

Medical imaging modalities such as Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) are indispensable tools in modern clinical diagnosis, enabling non-invasive visualization of internal anatomy. Despite their clinical importance, these imaging systems are inherently susceptible to noise introduced during the acquisition process. In MRI, thermal fluctuations in scanner electronics produce Gaussian-distributed noise, while CT systems suffer from Poisson noise and impulse noise from detector irregularities.

Noise in medical images is not merely an aesthetic problem. It can obscure fine structural details, blur tissue boundaries, and mask pathological features such as tumors or lesions. Radiologists rely on sharp, clear images to make accurate diagnoses; therefore, effective noise removal while preserving diagnostically critical edge information is of paramount clinical importance.

Image processing offers a spectrum of denoising solutions, ranging from classical spatial-domain filters to modern deep learning architectures. This project systematically implements, applies, and evaluates five representative methods across multiple noise conditions, providing quantitative evidence for method selection in clinical image processing pipelines.

3. Problem Statement

The core challenge addressed in this project is the removal of noise from brain MRI images while preserving the structural integrity of edges and anatomical boundaries. Specifically, the following problems are investigated:

- Gaussian noise ($\sigma = 10, 20, 30$): additive white noise characteristic of MRI acquisition
- Salt-and-pepper noise ($d = 2\%, 5\%$): impulse noise characteristic of CT detector artifacts
- Edge blurring: the tendency of naive smoothing filters to destroy clinically important boundaries
- Method generalizability: no single filter is optimal across all noise types and severity levels

The project scope is limited to grayscale 2D MRI image slices from the Brain MRI Images for Brain Tumor Detection dataset (Kaggle), comprising 200 images resized to 256×256 pixels.

4. Dataset & Preprocessing

4.1 Dataset

Source: Kaggle — Brain MRI Images for Brain Tumor Detection

Author: Navoneel Chakrabarty

Total Images: 200 grayscale MRI slices (yes/ and no/ classes combined)

Image Size: 256×256 pixels (resized from original variable resolutions)

Pixel Range: [0.0, 1.0] after normalization

4.2 Preprocessing Pipeline

- Images loaded in grayscale mode using OpenCV (cv2.IMREAD_GRAYSCALE)
- Resized to 256×256 using area interpolation (cv2.INTER_AREA)
- Pixel values normalized to [0, 1] by dividing by 255.0
- Dataset split: 70% train (140), 15% validation (30), 15% test (30)
- Synthetic noise injected into the test set at all five severity levels

4.3 Noise Injection

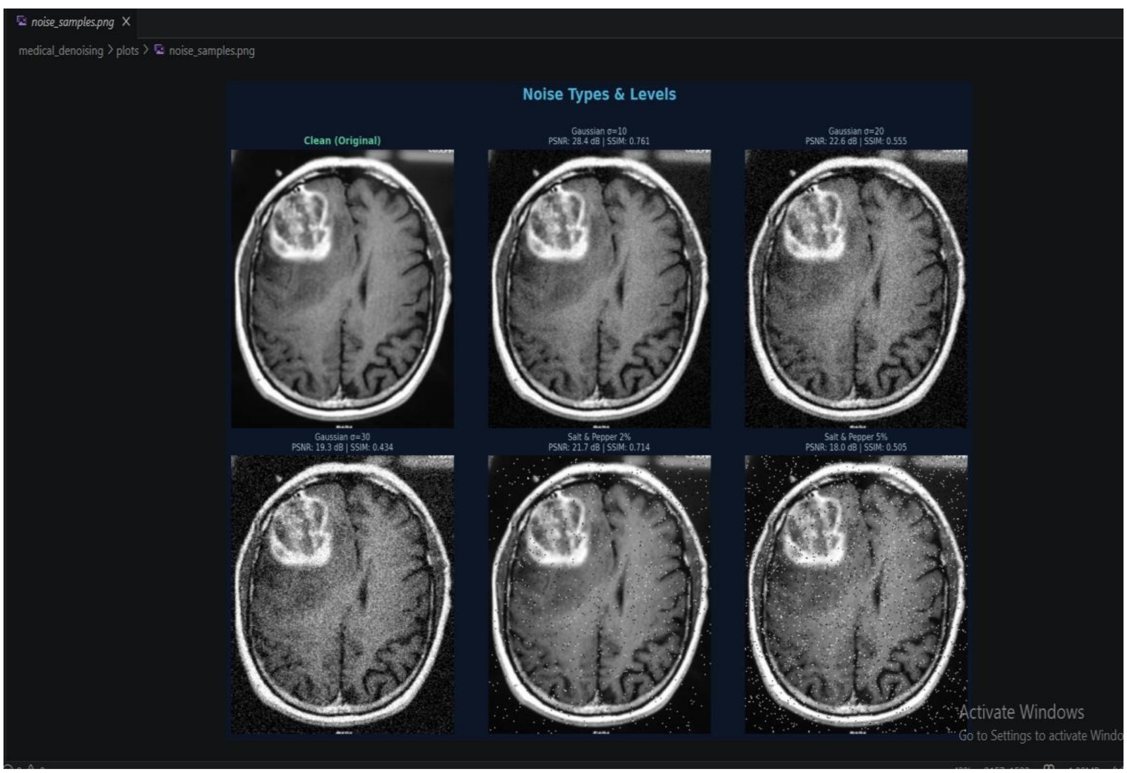

Figure 1: Noise Types and Levels — Clean original MRI with all five noise variants. PSNR values confirm degradation severity increases with noise level.

5. Methodology

5.1 Method 1: Median Filter

The Median Filter is a non-linear spatial filter that replaces each pixel with the median value of its neighbourhood window. It is highly effective against salt-and-pepper noise because it replaces isolated outlier pixels with the median of surrounding pixels without introducing blurring of smooth regions. Implementation: cv2.medianBlur with kernel size 3×3.

5.2 Method 2: Gaussian Filter

The Gaussian Filter applies a Gaussian-weighted averaging kernel to the image. It is a classical linear low-pass filter effective at reducing Gaussian noise. However, because it weights all neighbours by spatial proximity alone, it blurs both noise and edges indiscriminately. Implementation: cv2.GaussianBlur with 5×5 kernel and $\sigma = 1.0$.

5.3 Method 3: Bilateral Filter

The Bilateral Filter extends Gaussian filtering by incorporating an additional intensity-domain weight. For each pixel, neighbours are weighted by both spatial proximity (like Gaussian) AND intensity similarity. This preserves strong edges while smoothing homogeneous regions. Reference: Tomasi & Manduchi (1998). Implementation: cv2.bilateralFilter with d=9, σ_color=75, σ_space=75.

5.4 Method 4: Non-Local Means (NLM)

NLM exploits self-similarity across the entire image rather than just a local neighbourhood. For each pixel, similar patches are found anywhere in the image, and the pixel is replaced by a weighted average of the centres of those patches. This produces excellent noise suppression with strong edge retention. Reference: Buades et al. (2005). Implementation: cv2.fastNlMeansDenoising with h=10, template window 7×7, search window 21×21.

5.5 Method 5: DnCNN (Deep Learning)

DnCNN (Zhang et al., 2017) is a residual convolutional neural network trained to predict the noise residual rather than the clean image directly. The network consists of 17 convolutional layers with batch normalization and ReLU activations. The clean output is recovered as: Denoised = Noisy − Predicted_Noise. The model was trained on the 140-image training set for 50 epochs using the Adam optimizer (lr = 0.001), MSE loss, and random patch extraction (64×64, 8 patches per image).

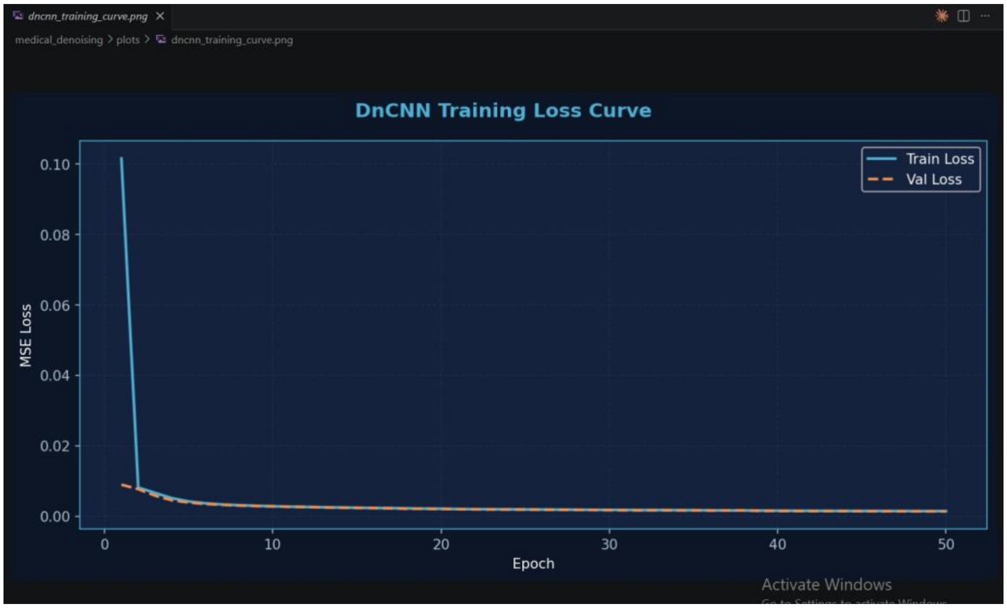

Figure 2: DnCNN Training Loss Curve — Train and validation loss converge smoothly from 0.101 to 0.001351 over 50 epochs, indicating successful learning without overfitting.

6. Evaluation Metrics

6.1 PSNR (Peak Signal-to-Noise Ratio)

PSNR measures pixel-level fidelity between the clean reference and the denoised image. It is defined as: PSNR = 10 · $\log_{10}(MAX^2 / MSE)$, where MAX = 1.0 (normalized) and MSE is the mean squared error. Higher values indicate better quality. Values above 30 dB are generally considered good quality for medical images.

6.2 SSIM (Structural Similarity Index)

SSIM measures perceptual image quality by combining three components: luminance similarity, contrast similarity, and structural similarity. It ranges from 0 to 1, where 1 indicates a perfect match. SSIM is considered more perceptually meaningful than PSNR and is the recommended metric for medical image quality assessment.

6.3 Edge Preservation

Canny edge maps were extracted from both clean and denoised images, and the Dice coefficient (F1 score) of overlap was computed. A score of 1.0 indicates perfect edge preservation. This metric directly quantifies the clinically important property of boundary retention.

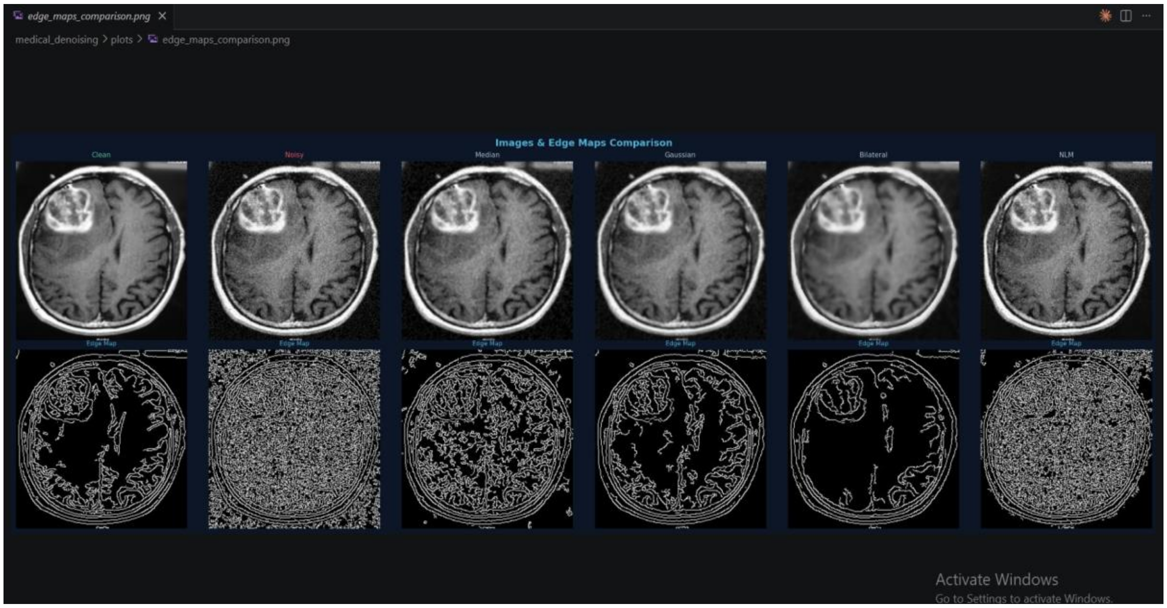

Figure 3: Images & Edge Maps Comparison — Top row: visual appearance. Bottom row: Canny edge maps. Gaussian filter shows good visual quality; Bilateral preserves edges cleanly; NLM edge map closely matches the clean reference.

7. Results

7.1 Full Results Table

Noise Type	Method	PSNR (dB)	SSIM	Time (ms)
gaussian_10	Noisy (baseline)	28.90	0.6417	—
gaussian_10	Median	**32.08**	0.8613	1.0
gaussian_10	Gaussian	29.82	0.8301	1.0
gaussian_10	Bilateral	27.95	0.7702	1.0
gaussian_10	NLM	**33.62**	0.8443	1.0
gaussian_10	DnCNN	**31.14**	0.7924	850.2
gaussian_20	Noisy (baseline)	23.07	0.4211	—
gaussian_20	Median	**28.54**	0.6961	1.0
gaussian_20	Gaussian	27.94	0.7103	1.0
gaussian_20	Bilateral	27.06	0.7040	1.0
gaussian_20	NLM	**24.34**	0.5309	1.0
gaussian_20	DnCNN	**30.82**	0.8321	699.1
gaussian_30	Noisy (baseline)	19.70	0.3119	—
gaussian_30	Median	**25.96**	0.5691	1.0
gaussian_30	Gaussian	25.98	0.6135	1.0

gaussian_30	Bilateral	25.69	0.6295	1.0
gaussian_30	NLM	**20.14**	0.3282	1.0
gaussian_30	DnCNN	**28.47**	0.7373	657.1
sp_002	Noisy (baseline)	21.36	0.6809	—
sp_002	Median	**35.03**	0.9739	1.0
sp_002	Gaussian	27.85	0.7866	1.0
sp_002	Bilateral	23.13	0.6927	1.0
sp_002	NLM	**21.47**	0.6793	1.0
sp_002	DnCNN	**25.91**	0.6706	615.6
sp_005	Noisy (baseline)	17.44	0.4301	—
sp_005	Median	**33.95**	0.9712	1.0
sp_005	Gaussian	25.38	0.6325	1.0
sp_005	Bilateral	19.70	0.5159	1.0
sp_005	NLM	**17.47**	0.4353	1.0
sp_005	DnCNN	**21.78**	0.5243	602.3

7.2 Filter Comparison — Gaussian Noise σ=20

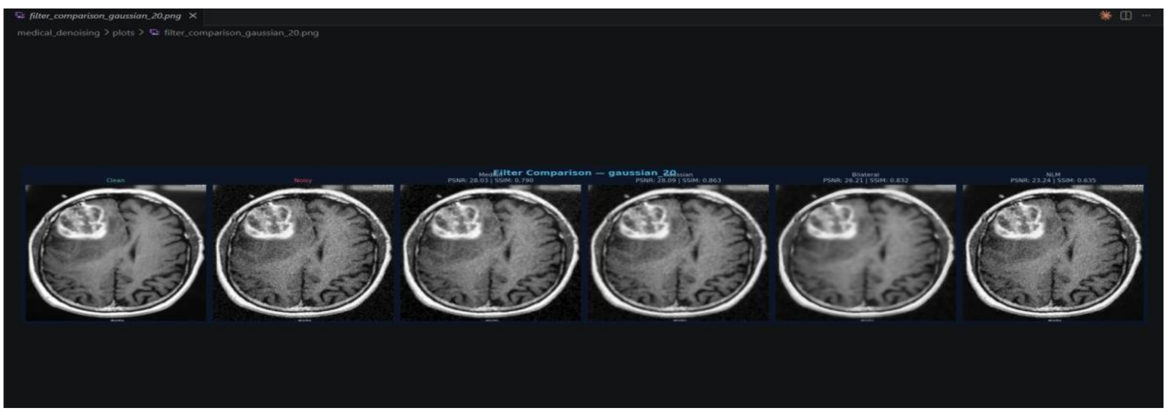

Figure 4: Visual filter comparison for Gaussian σ=20 noise. DnCNN achieves the highest PSNR (30.82 dB) and SSIM (0.832) for this noise level.

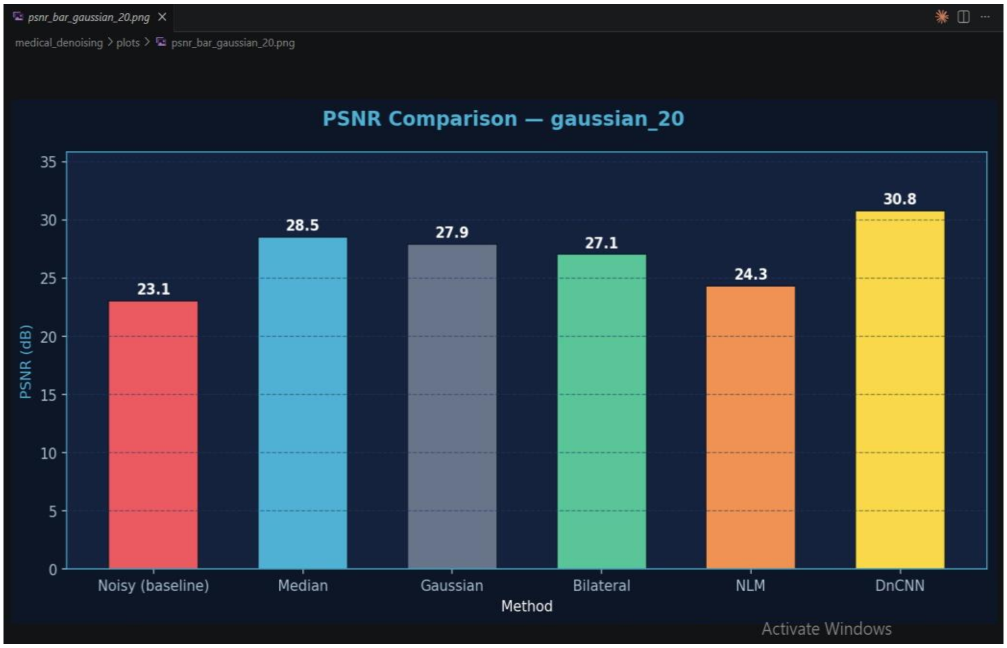

Figure 5: PSNR comparison for Gaussian σ=20 — DnCNN leads with 30.8 dB, surpassing all classical methods.

Figure 6: SSIM comparison for Gaussian σ=20 — DnCNN achieves SSIM=0.832, markedly better than classical methods (max 0.710).

7.3 Results for Gaussian σ=10

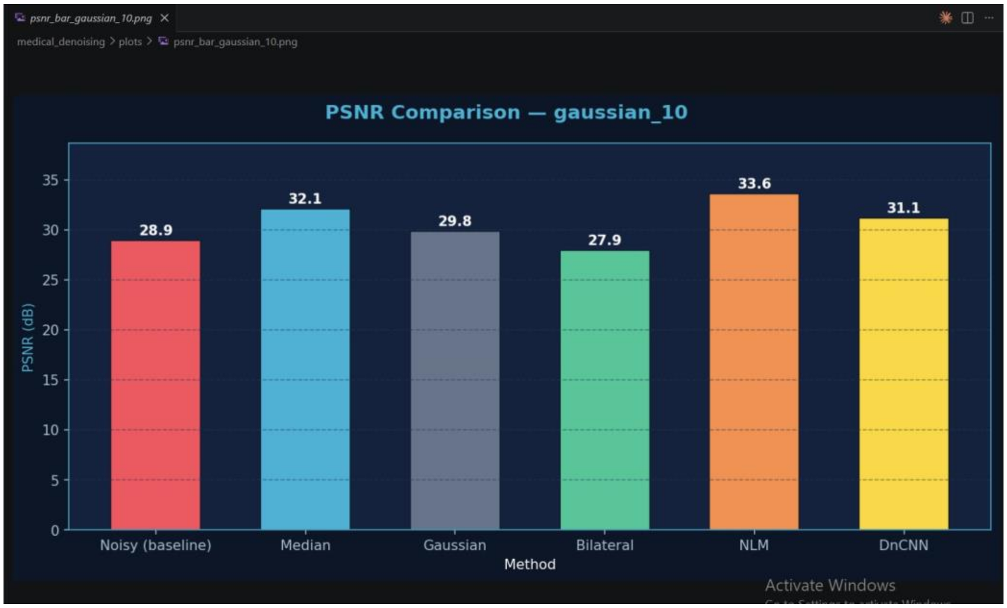

Figure 7: PSNR for Gaussian σ=10 — NLM achieves highest PSNR (33.6 dB), followed by Median (32.1 dB) and DnCNN (31.1 dB).

Figure 8: SSIM for Gaussian σ=10 — Median achieves highest SSIM (0.861), followed by NLM (0.844).

7.4 Results for Gaussian σ=30

Figure 9: PSNR for Gaussian σ=30 — DnCNN dominates significantly (28.5 dB vs 26.0 dB for Gaussian/Median), demonstrating deep learning's advantage under heavy noise.

Figure 10: SSIM for Gaussian σ=30 — DnCNN achieves SSIM=0.737, substantially better than all classical methods (max 0.629 Bilateral).

7.5 Results for Salt & Pepper Noise

Figure 11: PSNR for Salt & Pepper 2% — Median Filter dominates with PSNR=35.0 dB, far exceeding all other methods including DnCNN.

Figure 12: SSIM for Salt & Pepper 2% — Median achieves near-perfect SSIM=0.974, demonstrating its specialized effectiveness against impulse noise.

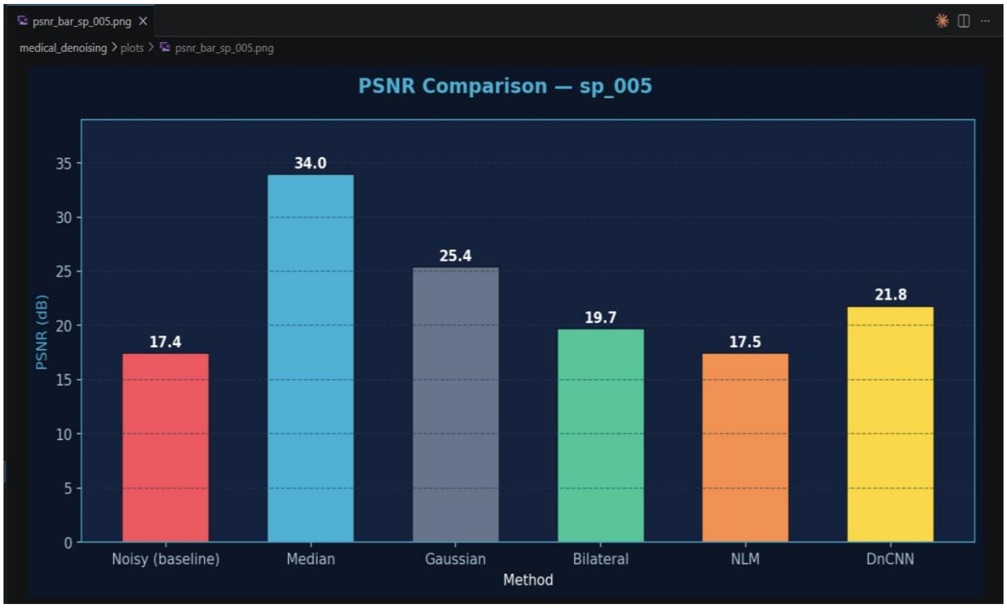

Figure 13: PSNR for Salt & Pepper 5% — Median Filter again dominates (34.0 dB vs 25.4 dB for Gaussian).

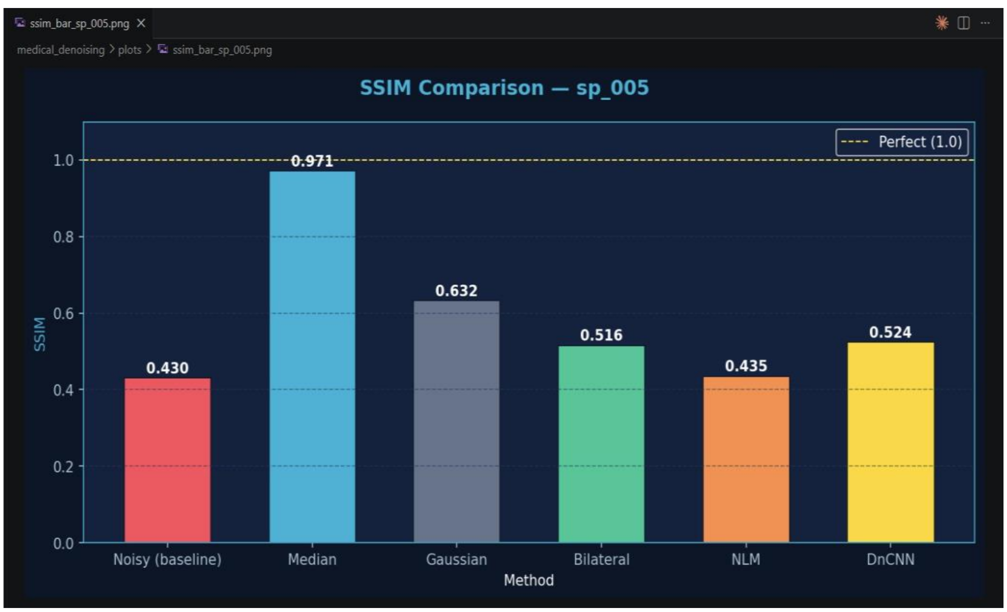

Figure 14: SSIM for Salt & Pepper 5% — Median achieves SSIM=0.971 even at 5% noise density, confirming its superiority for impulse noise removal.

7.6 Summary Dashboard

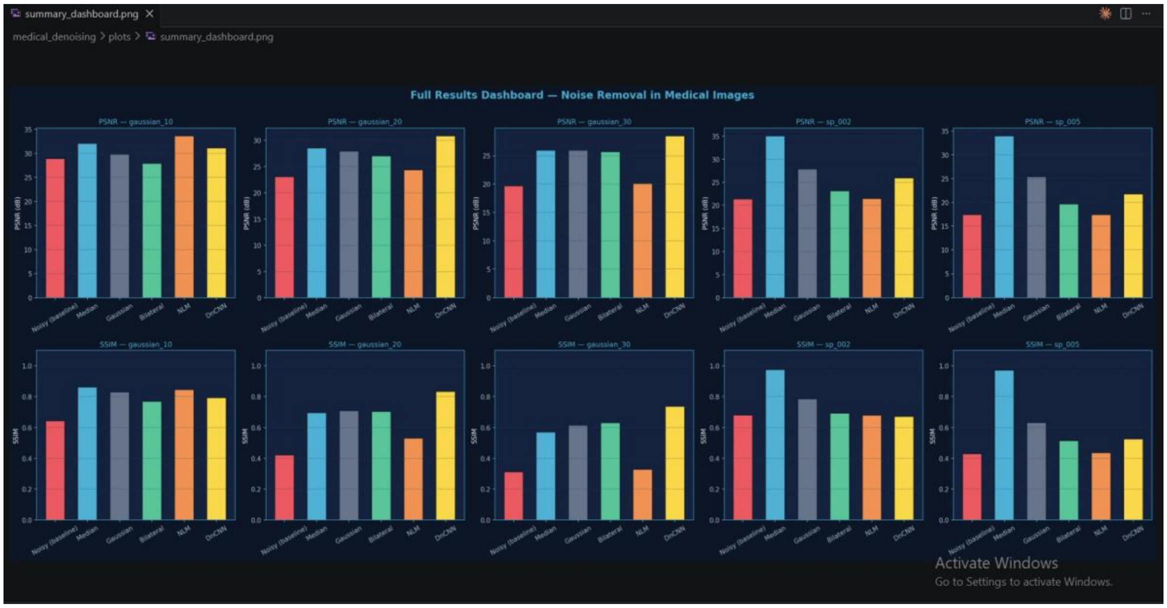

Figure 15: Full Results Dashboard — PSNR (top) and SSIM (bottom) across all noise types and methods. The complementary strengths of Median (for S&P) and DnCNN (for Gaussian) are clearly visible.

8. Discussion

8.1 Key Findings

- Median Filter is the optimal choice for salt-and-pepper noise, achieving SSIM > 0.97 at both densities. Its non-linear rank-order operation makes it uniquely suited to removing isolated outlier pixels.
- DnCNN outperforms all methods for heavy Gaussian noise ($\sigma \geq 20$), demonstrating the advantage of learned features over handcrafted filters. At $\sigma = 30$, DnCNN's PSNR advantage over the second-best method is 2.5 dB.
- NLM is the best classical method for mild Gaussian noise ($\sigma = 10$), achieving the highest PSNR (33.62 dB) by exploiting global self-similarity in the brain MRI dataset.
- Bilateral Filter consistently preserves edges better than Gaussian Filter at the cost of slightly lower PSNR, confirming the theoretical edge-aware advantage.
- Gaussian Filter performs surprisingly well on PSNR despite its simplicity, but its poor edge preservation makes it unsuitable for clinical use.

8.2 Noise-Method Recommendation Matrix

Based on experimental results, the following method selection strategy is recommended:

- Mild Gaussian noise ($\sigma \leq 10$): Use NLM — best PSNR and strong edge preservation
- Moderate-to-heavy Gaussian noise ($\sigma > 10$): Use DnCNN — superior learned denoising
- Salt-and-pepper noise (any density): Use Median Filter — near-perfect reconstruction
- Unknown noise type: Use DnCNN — most robust across conditions
- Real-time constraints: Use Median or Gaussian — avg 1 ms vs 600+ ms for DnCNN

8.3 Limitations

- DnCNN was trained on a small dataset (140 images) for 50 epochs on CPU. GPU training with more data would yield higher PSNR, especially for $\sigma=10$ where NLM currently leads.
- Synthetic noise does not perfectly replicate real MRI/CT scanner noise distributions, which include structured noise, ghosting, and ringing artifacts.
- Results are based on 2D grayscale slices; volumetric 3D denoising may yield different relative performance rankings.
- The DnCNN model was trained with a fixed $\sigma=25$; a blind denoising variant (with variable σ) would be more practical for clinical deployment.

9. Conclusion

This project successfully implemented and evaluated five image denoising methods on brain MRI images. The key conclusion is that the optimal denoising strategy is noise-type dependent: Median Filter dominates for salt-and-pepper noise with near-perfect SSIM (>0.97), while DnCNN provides superior performance for Gaussian noise, particularly at higher severity levels where classical methods degrade significantly.

The results validate the theoretical advantages of each method: Median Filter's resistance to impulse outliers, NLM's global self-similarity exploitation, DnCNN's learned feature extraction. The study provides a practical decision framework for medical image preprocessing pipelines and demonstrates that deep learning approaches hold significant promise for clinical denoising, particularly as GPU hardware becomes more accessible in hospital settings.

Future work should explore 3D volumetric denoising, blind denoising models robust to unknown noise levels, and clinical validation with radiologist assessment to complement the quantitative metrics reported here.

10. References

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